

ANISH NAIR

25070122037

A1

1.Method Overloading

A software company is developing a **calculator application** that should support addition of different types of inputs:

- Two integers
- Two floating-point numbers
- Three integer

2. Method Overriding:

A banking system has a base class BankAccount that provides a method to calculate the interest rate. Different types of bank accounts (e.g., SavingsAccount, CurrentAccount) have different interest rates, so they should override the method.

VS code Folder stuff > J Main.java

```
1  public class Main {
2
3      static int add(int a, int b) {
4          return a + b;
5      }
6
7      static double add(double a, double b) {
8          return a + b;
9      }
10
11     static int add(int a, int b, int c) {
12         return a + b + c;
13     }
14
15     public static void main(String[] args) {
16         System.out.println(add(5, 10));
17         System.out.println(add(5.5, 10.2));
18         System.out.println(add(5, 10, 15));
19     }
20 }
```

15

15.7

30

VS code Folder stuff > J Main.java

```
1  class BankAccount {
2      double calculateInterest() {
3          return 0.0;
4      }
5  }
6
7  class SavingsAccount extends BankAccount {
8      double calculateInterest() {
9          return 4.5;
10     }
11 }
12
13 class CurrentAccount extends BankAccount {
14     double calculateInterest() {
15         return 1.0;
16     }
17 }
18
19 public class BankTest {
20     public static void main(String[] args) {
21         BankAccount acc1 = new SavingsAccount();
22         BankAccount acc2 = new CurrentAccount();
23
24         System.out.println(acc1.calculateInterest());
25         System.out.println(acc2.calculateInterest());
26     }
27 }
```

4.5

1.0